Chapter 1 exercises: ADTs, correctness, and cost

DS01-E01

A stack of capacity 3 receives push(2), push(2), push(5), pop, push(7), push(8). A successful push returns success; a full push returns failure and leaves the state unchanged. A nonempty pop returns the removed element. Give each state and return value. Why is a set inadequate as the abstract state?

DS01-E02

Design a dynamically allocated stack push with the strong exception guarantee. State the ordering of allocation, construction, commitment, and cleanup, and the necessary assumptions about element copying or moving.

DS01-E03

Trace lower_bound on A=[1,2,2,2,5], x=2 using [l,r)=[0,5). Give the result and compare it with the algorithms course' s 1-based FIRST-MATCH, including absent-key behavior.

DS01-E04

For n=6 and each i=1,…,n, initialize x=n and execute $x \leftarrow x-i$ while $x \geq 0$. Count body executions and condition tests by i, including the final failed test.

DS01-E05

With n=8, halving until n=1, compare `2*Test(n/2)` and `Test(n/2)+Test(n/2)`: count all calls and derive each recurrence and asymptotic running time. Use a source-level model without common-subexpression elimination.

DS01-E06

For direct recursion with F(0)=0, F(1)=1, find F(5), call count C(5), and prove $C(n)=2F(n+1)-1$. Find G(5) for G(0)=G(1)=1, and determine whether its call count changes.

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DS01-E07

An empty dynamic array starts at capacity 3 and doubles when full. For 13 appends, list the triggering append, copies per resize, and total writes. For any initial capacity $c_0 \geq 1$, prove that copies for $n \geq 1$ appends are fewer than $2n$ and handle $n=0$. Exclude initial allocation.

DS01-E08

For $\Phi(s,C)=\max(0,2s-C)$, calculate actual and amortized cost of appending when $s=C=4$. Bound the amortized cost without a resize and explain the bound over a sequence.

DS01-E09

A four-bit counter charges one per bit flipped. With potential equal to the number of one bits, calculate actual and amortized costs for $0111 \to 1000$ and $1111 \to 0000$. Does the bound imply each actual operation costs at most two?

DS01-E10

For n equal keys, two-way Lomuto partition sends equals to one side. Give subproblem sizes and comparisons. Does a random pivot help? State the recursive rule of three-way partitioning.

DS01-E11

Prove $3n+8 \in O(n)$ with explicit constants. Decide whether $4^n \in O(2^n)$, with proof. Distinguish worst-case, expected, and amortized costs.

DS01-E12

For lower_bound on a random-access array of strings of length at most L, with lexicographic comparisons costing up to O(L) character operations, bound comparisons and character operations. State the n=0 result and explain the assumption behind constant-cost comparison.

DS01-E13 Bit cost of addition

Sequentially add n signed integers. Both operands and the result of each addition fit in at most L bits, and addition is performed bit by bit. Bound the total bit operations. If the final result is 0, does that imply constant time per addition? Explain using $2^N + (-2^N)$ and state when a constant-time fixed-word model is appropriate.

References and attribution

Lin Cing, handwritten notes on data structures.

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